

CSc 453: Fall 2025

Assignment 1 Milestone 1

Start: Sept 4, 2025

Due: 11:59 PM, Sept 10, 2025

1. General

This assignment involves writing a scanner for the C++ language. This scanner will then be used for the remaining assignments in the semester project.

2. Required functionality

2.1. Programming requirements

Your program should implement a function `get_token()` that behaves as follows:

- It reads input from **stdin**.
- It discards comments and whitespace (however, even though comments and whitespace are discarded, they are significant in that they signal the end of any previous token). The syntax for comments is given in Section 1.3 of the [GO language spec](#).
- It identifies and returns tokens based on the tokens and patterns given in Section 1.2 of the [GO language spec](#).

Your program should use the following two files:

- [scanner.h](#) -- a header file that defines the set of tokens together with their values; and
- [scanner-driver.c](#) -- a file containing the driver code that invokes your scanner and prints out the results.

In subsequent assignments we will replace the driver file **scanner-driver.c** with a different driver, so you may not want to put your scanner implementation in this file.

You should submit a Makefile that provides (at least) the following targets:

- **make clean:**
Deletes any object files (*.o) as well as the file 'scanner'
- **make scanner:**
Compiles all the files from scratch and creates an executable file named 'scanner'.

2.2. Interface requirements

Your program should be written to interface appropriately with the driver code mentioned above. To this end, your code should provide the following:

```
int get_token(); /* the scanner function */
char* lexeme;    /* a pointer to the current lexeme */
```

3. Files you need to submit

- Makefile that supports the functionality described above;
- scanner.h and scanner-driver.c -- the files provided to you (see links above);
- any additional files implementing your scanner.

4. How to submit your code

You should submit your code on [Gradescope](#):

- Click on the link "Assignment 1 milestone 1"
- Drag and drop all the files you want to submit into the "Submit Programming Assignment" window that comes up; or else click on this window to browse and select the files to submit. Click on the "Upload" button when ready.

Your code will be run by an autograder when you upload it into Gradescope as described above. This usually takes a couple of minutes. Once the autograder has run, you will get the following feedback:

- Before the submission deadline, you will get feedback on a single test file (to expose any issues or problems that may exist) together with a summary of how your code has done overall in terms of the number of test inputs it has passed.
- Once the submission deadline has passed, you will see detailed feedback on how your code performed on all the test inputs.

5. Common issues to avoid

Last-minute submissions

Please note that the submission deadline is 11:59 PM on the due date according to Gradescope's server's clock. This may or may not be the same as your clock. I recommend not waiting to find out.

Make

We often write make files so that typing just "make" builds the executable we want. However, the assignment requires that your make file should support the command "make scanner", which is the command the grading script uses to build the executable. Please check your make file before you submit.